

Practice Quiz: The Prototype as a System

Part A: Multiple Choice

1. A prototype functions as a physical generative model primarily because it: A) Looks like the final product B) Embodies the inventor's hypotheses and generates testable predictions C) Is made from the same materials as the final product D) Has been approved by potential customers
2. The Markov blanket of a prototype separates: A) The inventor from the user B) The prototype's internal mechanisms from its external testing environment C) Good prototypes from bad prototypes D) Physical prototypes from digital prototypes
3. A low-fidelity prototype is most useful when: A) The inventor has high confidence in most design parameters B) The invention is nearly ready for production C) The inventor's uncertainty is broad and many hypotheses need cheap, rapid testing D) The final product must be demonstrated to investors
4. The "kitchen sink prototype" fails because: A) It is too cheap to be taken seriously B) It includes too many variables, making it impossible to isolate which caused a given test result C) It does not include enough features D) It takes too long to build
5. When a pharmaceutical company tests a drug molecule in cell cultures before animal trials, they are applying the principle of: A) High-fidelity prototyping B) Progressive Markov blanket expansion — starting with a minimal boundary and expanding as uncertainty decreases C) Eliminating all external states from the system D) Skipping the prototyping phase entirely
6. A "Wizard of Oz" prototype replaces the system's internal states with a human operator while keeping the sensory and active surfaces intact. This is useful because: A) It eliminates the need for a prototype entirely B) It allows testing user interaction hypotheses without building the internal mechanism C) Human operators are always better than machines D) It reduces the prototype's Markov blanket to zero

7. In choosing between prototype fidelity levels, the expected free energy framework recommends: A) Always building the highest fidelity possible B) Always building the lowest fidelity possible C) Choosing the fidelity that maximizes information gain relative to cost and current uncertainty D) Matching the fidelity of competitors' prototypes

Part B: Short Answer

1. An inventor is developing a new ergonomic computer mouse. They are considering two prototype approaches: (a) a 3D-printed shell with no electronics, tested for hand comfort, or (b) a fully functional mouse with custom electronics, tested for both comfort and tracking precision. Using the concept of Markov blankets and fidelity levels, advise them on which approach to use first and why.
2. Explain why prototype design and test design are "two aspects of the same act of inference." Give a specific example where designing the prototype's sensory surface determines what can be learned from testing.
3. James Dyson built 5,127 prototypes of his cyclone vacuum. From an Active Inference perspective, why was it more effective to build thousands of simple prototypes than to build a few complex ones? Connect your answer to the concepts of model resolution, prediction error, and parameter updating.